Geiger-Marsden-Rutherford experiment

Most alpha particles pass through gold foil with no change in direction or energy loss. A small number of particles deviate from their original direction by angles more than 90°.

Some alpha particles are **deflected** because they come into very close proximity of the gold nucleus. Both the **gold nucleus** and the **alpha particles** are **positively charged**, hence there is a repulsive force between them.

The force varies as $\frac{1}{r^2}$ where r is the distance between the **centres** of the alpha particle and the gold nucleus.

Head-on scattering = angle of deflection is 180° = the alpha particles are deflected so that they return along the same path by which they arrived.

With head-on scattering, estimation for the radius of the scattering nucleus can be done.

Strictly when the alpha particle is stationary && head-on scattering, then kinetic energy E_α is transferred to the electric potential energy E_p stored **in the nucleus-alpha system**.

$$\therefore E_\alpha = E_p$$
$$\frac{1}{2} m_\alpha {v_\alpha}^2 = k \frac{(+2e)(+Ze)}{r}$$

Z information here.

Principal assumption: $\exists$ a repulsive force between the nucleus and alpha particle $\implies$ Coulomb's law applied to the interaction.

What we now know: $\exists$ an attractive strong nuclear interaction that acts between the alpha particle and the nucleus.